

```

1 • use mavenmovies;
2 • select * from film;
3
4 /*Business Question:
5 Findout 3 film category generates the most total revenue? */
6
7 • SELECT cat.name AS category, SUM(p.amount) AS total_revenue
8 FROM payment p
9 JOIN rental r ON p.rental_id = r.rental_id
10 JOIN inventory i ON r.inventory_id = i.inventory_id
11 JOIN film_category fc ON i.film_id = fc.film_id
12 JOIN category cat ON fc.category_id = cat.category_id
13 GROUP BY cat.name
14 ORDER BY total_revenue DESC
15 LIMIT 3;
16

```

category	total_revenue
Sports	5314.21
Sci-Fi	4756.98
Animation	4656.30

customer_id	first_name	last_name	total_spent	avg_payment
526	KARL	SEAL	221.55	4.923333
148	ELEANOR	HUNT	216.54	4.707391
144	CLARA	SHAW	195.58	4.656667
178	MARION	SNYDER	194.61	4.990000
137	RHONDA	KENNEDY	194.61	4.990000

staff_id	first_name	last_name	payments_processed	revenue_generated
1	Mike	Hillyer	8057	33489.47
2	Jon	Stephens	7992	33927.04

```
39
40 /*Business Question:
41 Find the busiest month for rentals (highest number of rentals) */
42
43
44 • SELECT DATE_FORMAT(rental_date, '%Y-%m') AS rental_month,
45 COUNT(*) AS rental_count
46 FROM rental
47 GROUP BY rental_month
48 ORDER BY rental_count DESC
49 LIMIT 1;
50
```

create_mavenmovies* Business_Analytics_Queries* x

Limit to 50000 rows

```

51 /*Business Question:
52 Find films that exist in inventory but have never been rented (unused stock) */
53
54 • SELECT f.film_id,f.title,
55 COUNT(i.inventory_id) AS unused_copies
56 FROM film f
57 JOIN inventory i ON f.film_id = i.film_id
58 LEFT JOIN rental r ON i.inventory_id = r.inventory_id
59 WHERE r.rental_id IS NULL
60 GROUP BY f.film_id, f.title
61 ORDER BY unused_copies DESC;
62

```

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: |

	film_id	title	unused_copies
▶ 1	ACADEMY DINOSAUR		1

create_mavenmovies* Business_Analytics_Queries* x

Limit to 50000 rows

```

63 /*Business Question:
64 Calculate, for each country, the number of customers and total revenue generated */
65
66 • SELECT co.country,
67 COUNT(DISTINCT c.customer_id) AS num_customers,
68 ROUND(SUM(p.amount),2) AS total_revenue
69 FROM country co
70 JOIN city ci ON co.country_id = ci.country_id
71 JOIN address a ON ci.city_id = a.city_id
72 JOIN customer c ON a.address_id = c.address_id
73 JOIN payment p ON c.customer_id = p.customer_id
74 GROUP BY co.country
75 ORDER BY total_revenue DESC;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: |

	country	num_customers	total_revenue
▶	India	60	6630.27
	China	53	5802.73
	United States	36	4110.32
	Japan	31	3471.74
	Mexico	30	3307.04
	Brazil	28	3200.52
	Russian Federation	28	3045.87
	Philippines	20	2381.32
	Turkey	15	1662.12
	Nigeria	13	1511.48
	Indonesia	14	1510.33

77  /*Business Question:

78 Find the average number of rentals per active customer vs. inactive customer */

79

80 • **SELECT** c.active,

81 ROUND(COUNT(r.rental_id)*1.0 /COUNT(DISTINCT c.customer_id),2) **AS** avg_rentals_per_customer

82 **FROM** customer c

83 **LEFT JOIN** rental r **ON** c.customer_id = r.customer_id

84 **GROUP BY** c.active;

85

Result Grid   Filter Rows: | Export:  Wrap Cell Content: 

	active	avg_rentals_per_customer
0		26.93
1		26.78

create_mavenmovies* Business_Analytics_Queries* x

Limit to 50000 rows

```
86 /*Business Question:
87 Find the actor who has appeared in the most films, along with the count */
88
89 • SELECT a.actor_id, a.first_name, a.last_name, COUNT(fa.film_id) AS film_count
90 FROM actor a
91 JOIN film_actor fa ON a.actor_id = fa.actor_id
92 GROUP BY a.actor_id, a.first_name, a.last_name
93 ORDER BY film_count DESC
94 LIMIT 1;
95
96 /*Business Question:
```

Result Grid

	actor_id	first_name	last_name	film_count
▶	107	GINA	DEGENERES	42

create_mavenmovies* Business_Analytics_Queries* x

Limit to 50000 rows

```
96 /*Business Question:
97 Using a CTE and window function, find each customer's first and most recent rental date
98 and total rentals in one query (cohort-style analysis).*/
99
100 • WITH cust_rentals AS (
101 SELECT customer_id,
102 MIN(rental_date) AS first_rental,
103 MAX(rental_date) AS last_rental,
104 COUNT(*) AS total_rentals
105 FROM rental
106 GROUP BY customer_id
107 )
108 SELECT c.first_name, c.last_name, cr.first_rental, cr.last_rental, cr.total_rentals,
109 DATEDIFF(cr.last_rental, cr.first_rental) AS days_active
110 FROM cust_rentals cr
111 JOIN customer c ON c.customer_id = cr.customer_id
112 ORDER BY total_rentals DESC;
113
```

Result Grid

	actor_id	first_name	last_name	film_count
▶	107	GINA	DEGENERES	42

create_mavenmovies*
Business_Analytics_Queries*

Limit to 50000 rows

```

113
114
115 /*Business Question:
116 Build a query that flags customers as "At Risk" if their last rental
117 was more than 60 days before the most recent rental date in the
118 whole dataset (churn analysis). */
119 WITH last_global AS (
120     SELECT MAX(rental_date) AS max_date FROM rental
121 ),
122 cust_last AS (
123     SELECT customer_id, MAX(rental_date) AS last_rental
124     FROM rental
125     GROUP BY customer_id
126 )
127 SELECT c.customer_id, c.first_name, c.last_name, cl.last_rental,
128 CASE
129     WHEN DATEDIFF(lg.max_date, cl.last_rental) > 60 THEN 'At Risk'
130     ELSE 'Active'
131 END AS status
132 FROM cust_last cl
133 JOIN customer c ON c.customer_id = cl.customer_id
134 CROSS JOIN last_global lg
135 ORDER BY status, cl.last_rental;

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content: IA

	customer_id	first_name	last_name	last_rental	status
▶	75	TAMMY	SANDERS	2006-02-14 15:16:03	Active
	335	GREGORY	MAULDIN	2006-02-14 15:16:03	Active
	317	EDWARD	BAUGH	2006-02-14 15:16:03	Active
	215	JESSIE	BANKS	2006-02-14 15:16:03	Active
	5	ELIZABETH	BROWN	2006-02-14 15:16:03	Active
	315	KENNETH	GOODEN	2006-02-14 15:16:03	Active
	263	HILDA	HOPKINS	2006-02-14 15:16:03	Active